ELYSIUM Mars Mission Architecture

TECHNICAL SPECIFICATION & MAP ENGINE BLUEPRINT • VERSION 1.0

ELYSIUM is a hybrid planetary GIS and AI-driven mission planning system designed to simulate surface exploration, traverse routing, and life-support constraints on Mars. By uniting high-resolution orbital raster streams with deterministic physics models, ELYSIUM provides astronauts and mission operators with real-time terrain intelligence.

Frontend Visualization

Leaflet 2D multi-layer canvas synced with a Three.js 3D planetary globe and dynamic local terrain mesh generator.

AI & Routing Engine

Weighted A* graph search over multi-layered rasters evaluating slope, slip, radiation, and consumables in real time.

Core Map Engine Workflow

Stage	Subsystem Action	Data Payload / Output
1. User Input	Captures start/end coordinates or freehand custom polylines drawn on Leaflet/Three.js.	GeoJSON coordinate array (`[lat, lon]`)
2. Raster Query	Extracts spatial bounding box tensors from local spatial database (DEM, Thermal Inertia, AI4Mars).	$128 \times 128 \times 5$ multi-band tensor
3. Physics Engine	Computes wheel slip %, climb energy (kWh), oxygen consumption (kg), and radiation dose (mSv).	Node-by-node segment telemetry
4. Dual-Channel Output	Responds with natural-language assessment, comparative viability score, and 3D tube geometries.	Unified JSON (HUD Text + 3D Arrays)

Mission Planning Capabilities

The system goes beyond basic geometric routing by embedding operational constraints faced during actual planetary rover and crewed deployments:

1. Custom Route Auditing

Users can sketch custom paths. The AI audits every 100m segment, highlights hazardous slopes (>15°) and sand traps in red, and outputs a 0–100% Viability Score.

2. Sol-by-Sol Timeline

Breaks traverses into daily Sol milestones, tracking battery state-of-charge, nighttime survival heater drain, and daily solar recharging.

3. Consumables & O2 Budget

Calculates metabolic and rover oxygen consumption relative to travel duration, factoring in winch resistance and soil sinkage.

4. Environmental Stress-Testing

Simulates regional dust storm opacity (τ) drops on solar arrays and solar flare (SPE) radiation alarms routing to cave shelters.

Routing Optimization Profiles

Route Strategy	Primary Objective	Cost Function Weights
Fastest Route	Minimizes straight-line distance; skirts only lethal cliffs and deep dunes.	Distance ($w_d = 0.85$), Slope ($w_s = 0.15$)
Safest Route	Heavily penalizes slope, wheel slip, boulder fields, and high radiation.	Slope ($w_s = 0.55$), Radiation ($w_r = 0.30$)
Optimum Route	Balances energy efficiency, safety, and science/water-ice reward proximity.	Energy ($w_e = 0.30$), Science ($w_{\text{sci}} = 0.15$)

Master Dataset Inventory (28 Core NASA / USGS Datasets)

The ELYSIUM backend indexes and ingests the following curated planetary datasets to feed its spatial pathfinding and machine learning pipelines:

#	Dataset Name	Mission / Agency	Primary AI Feature
1	MOLA / HRSC Blended Global DEM (200m)	Mars Express / MGS	Elevation (z) & Slope (θ)
2	HiRISE Digital Terrain Models (1m/px)	MRO HiRISE	Meter-scale obstacle avoidance
3	MOLA Vertical Roughness Map	MGS Altimeter	Baseline RMS surface roughness
4	USGS/IAU Gazetteer of Planetary Nomenclature	USGS Astrogeology	Feature names & basin vectors
5	MGS TES Global Rock Abundance Map	Mars Global Surveyor	Cumulative Fractional Area (CFA %)
6	MGD3 Global Dune Database	THEMIS / HiRISE	Active sand dune hazard polygons
7	Robbins Global Impact Crater Database	MOLA / THEMIS	Rim heights & wall slope geometry
8	AI4Mars Rover Terrain Segmentation Labels	MSL / M2020	Soil, sand & rock segmentation
9	RSL & Active Slope Streak Catalog	MRO HiRISE	Dry-granular mass wasting points
10	SWIM Subsurface Water Ice Mapping Grids	Multi-Mission	Shallow water-ice probability
11	MRO SHARAD Subsurface Radargrams	MRO SHARAD	Buried glacier depth (m)
12	Odyssey GRS / HEND Hydrogen (WEH)	2001 Mars Odyssey	Topsoil bound water/ice %
13	Mars Express MARSIS Soundings	Mars Express	Deep ionospheric & brine profiles
14	MDSSD Global Dust Storm Sequences	MRO MARCI	Seasonal dust storm corridors
15	MRO MCS Vertical Atmosphere Profiles	MRO MCS	Altitude density & opacity (τ)
16	MAVEN Solar Energetic Particle (SEP) Logs	MAVEN Orbiter	Acute solar storm radiation flux
17	MGS TES Global Bolometric Albedo Map	Mars Global Surveyor	Fine dust accumulation traps
18	MGC3 Global Cave & Lava Tube Catalog	THEMIS / HiRISE	Emergency radiation safe-havens
19	InSight SEIS Marsquake & Fault Catalog	InSight Lander	Tectonic epicenter & fault lines
20	NAIF SPICE Orbital Ephemeris Kernels	NASA NAIF	UHF/X-band relay line-of-sight
21	MRO / MGS JGMRO_120d Gravity Field	MRO Radio Science	Local gravity anomalies (g)
22	MRO CRISM Multispectral Mineralogy	MRO CRISM	Hydrated clays & sulfates